Domain-Driven Design (DDD) Architecture Report

Project Name: DRAKZ - Next-Generation Modular Financial Ecosystem

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1. Project Overview

Title:

DRAKZ: Next-Generation Modular Financial Ecosystem

Brief Description:

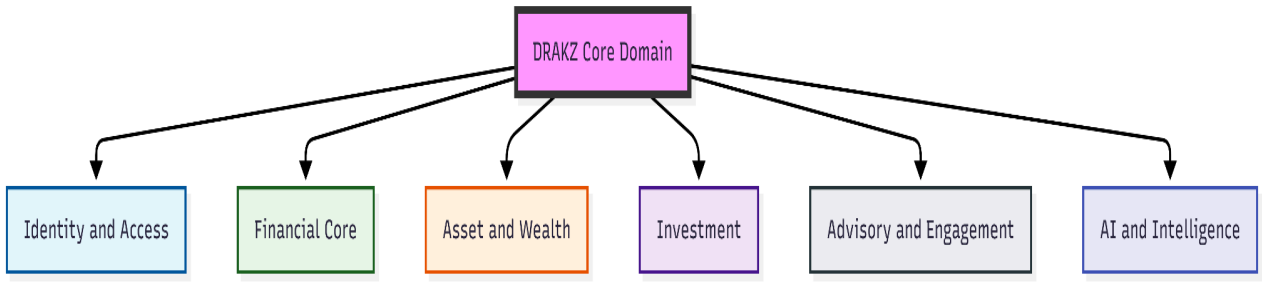
DRAKZ is a comprehensive, sophisticated financial management platform designed to provide users with a centralized hub for unified personal finance, wealth management, and expert advisory services. Built upon a modular, domain-driven architecture, the ecosystem seamlessly integrates day-to-day transaction tracking with long-term asset management covering bank accounts, expenses, stocks, mutual funds, real estate, and insurances.
Beyond standard tracking, DRAKZ differentiates itself by incorporating an AI-driven advisory layer and a dedicated Advisor Hub. Users interact with an intelligent financial chatbot for automated, actionable insights based on their spending and investment behaviors. Concurrently, the platform bridges the gap between automation and human expertise by allowing users to schedule secure consultation sessions with certified financial advisors. Advisors have access to dedicated CRM dashboards to track client portfolios and generate detailed financial reports. With its secure role-based infrastructure serving distinct User, Advisor, and Admin modules, DRAKZ elevates traditional personal finance by prioritizing intelligent decision-making, cohesive portfolio visualization, and real-time expert collaboration.

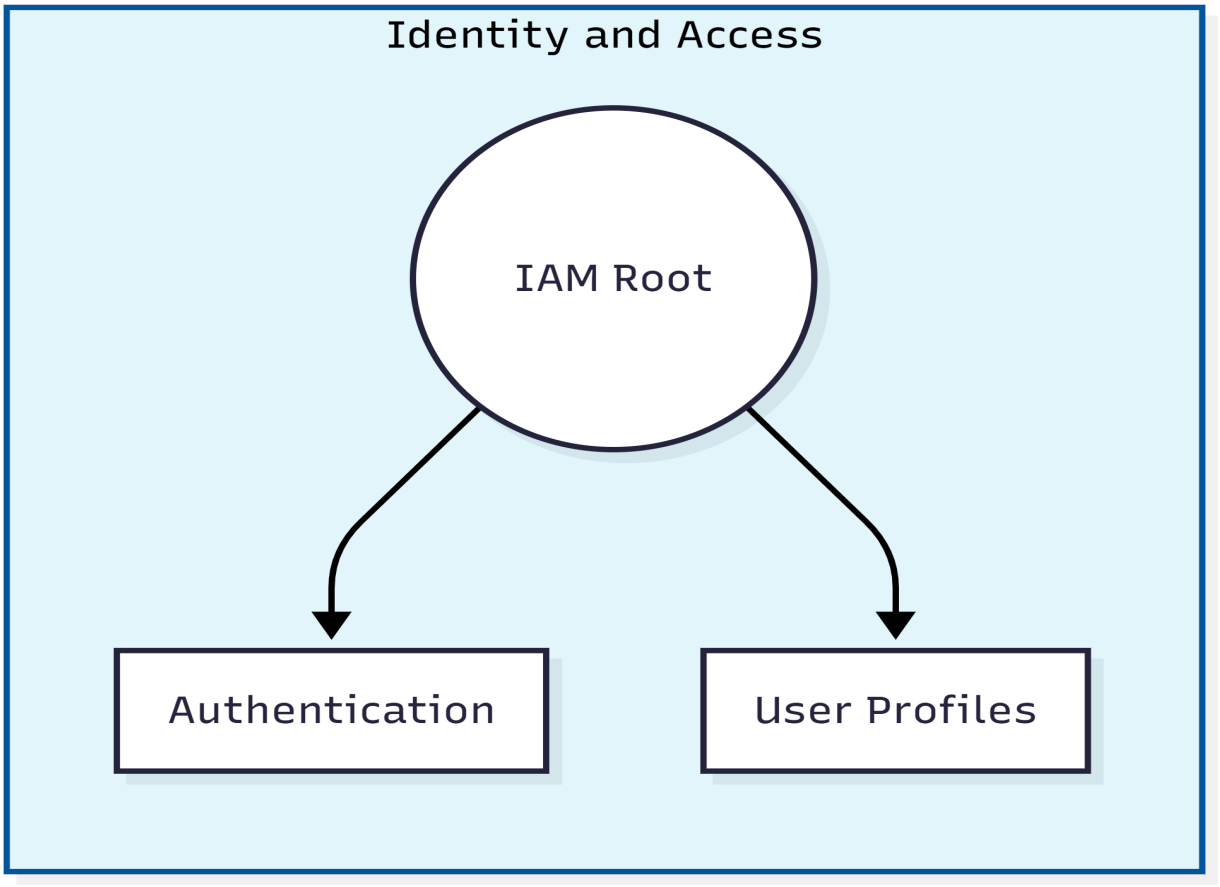
2. Domain-Driven Design (DDD) Specifications

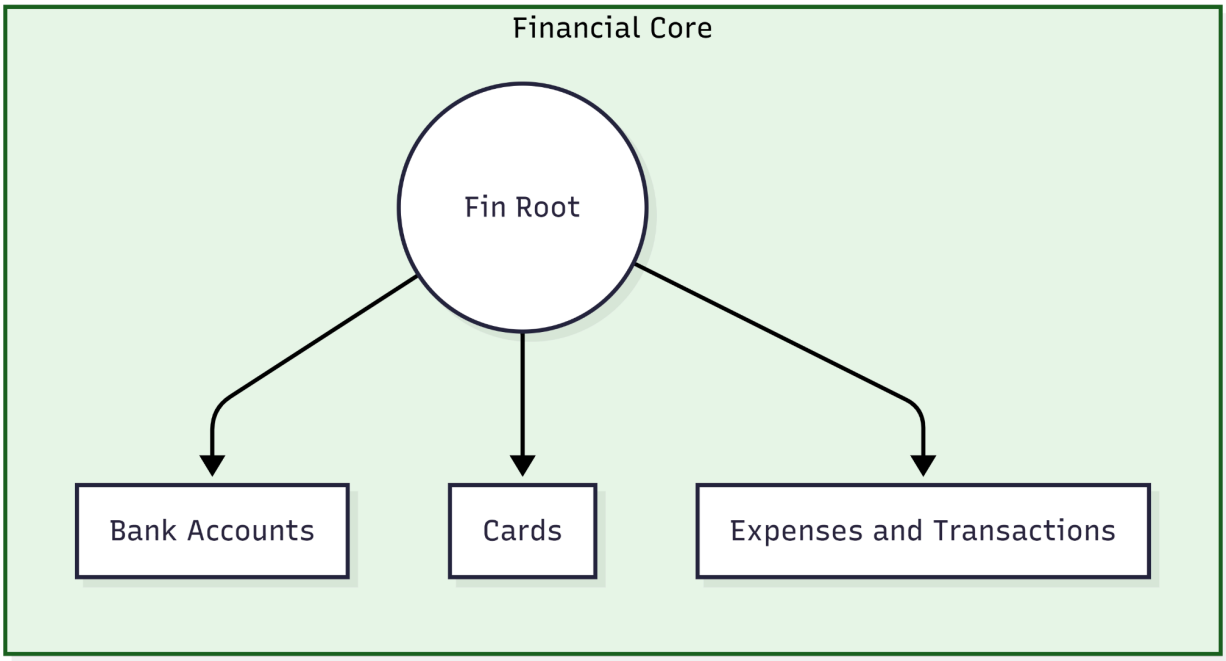
A. Bounded Contexts

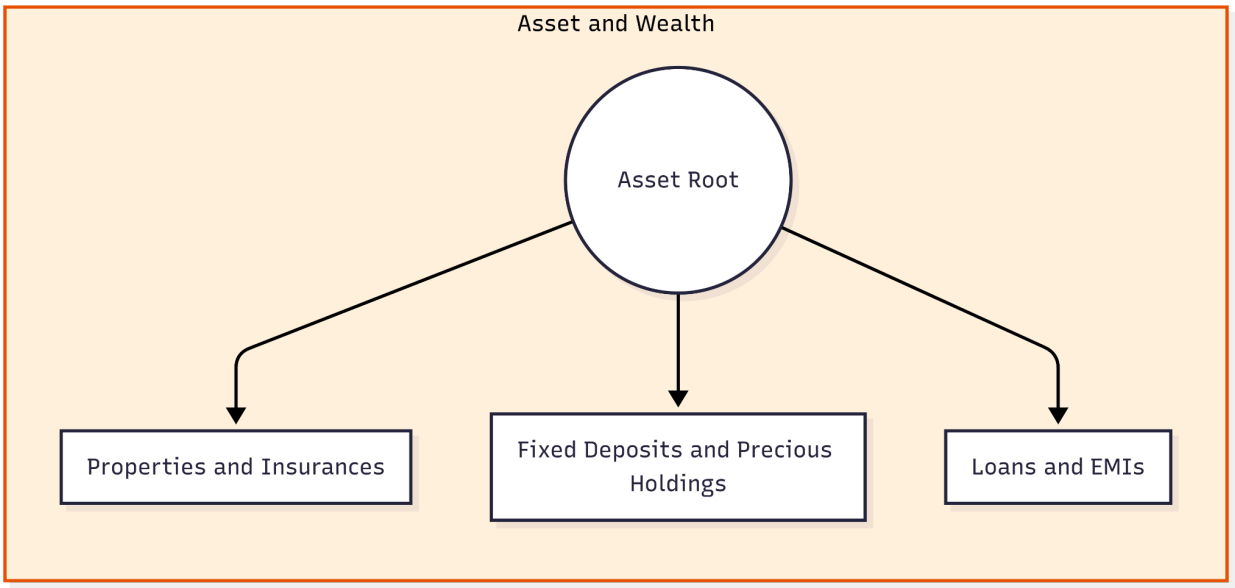
To maintain high cohesion and low coupling across a complex system, the DRAKZ Core Domain has been partitioned into six principal Bounded Contexts. Each context encapsulates its own ubiquitous language and business logic aligning strictly with the system's database schema.

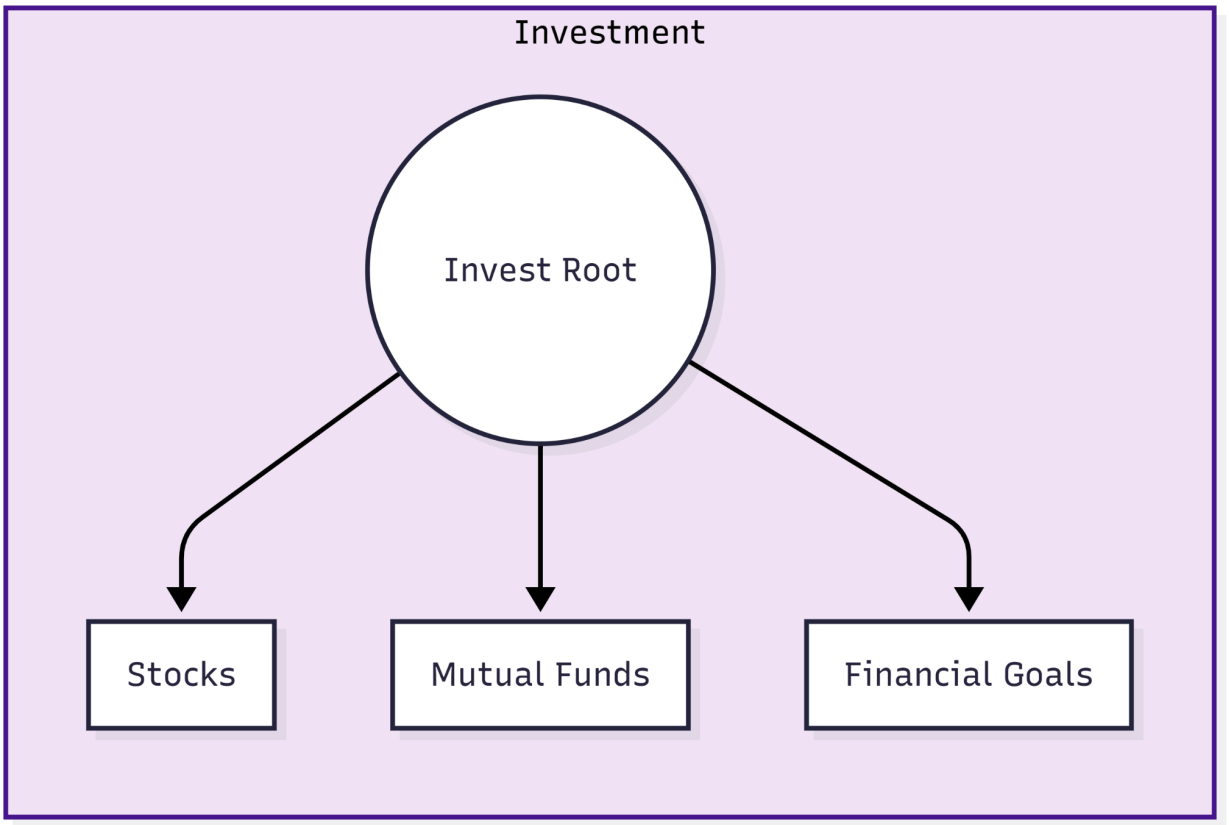
- Financial Core Context: Handles high-volume, day-to-day financial operations. It is bounded strictly to tracking operational cash flow rather than long-term asset valuation.
- Asset and Wealth Context: Specialized in tracking illiquid and semi-liquid assets such as physical properties, life insurances, precious holdings and fixed deposits.
- Investment Context: Manages market-driven, volatile financial instruments. It handles real-time portfolio tracking for stocks and mutual funds against specific financial goals.
- Advisory and Engagement Context: Facilitates B2B and B2C interactions, allowing certified advisors to manage authorized client portfolios, conduct video sessions, and publish financial articles.
- AI and Intelligence Context: Dedicated to processing natural language queries, running prediction models, and generating automated advisory outputs via the internal LLM orchestration layer. This is where the DRAKZ-1 chatbot middleware operates.

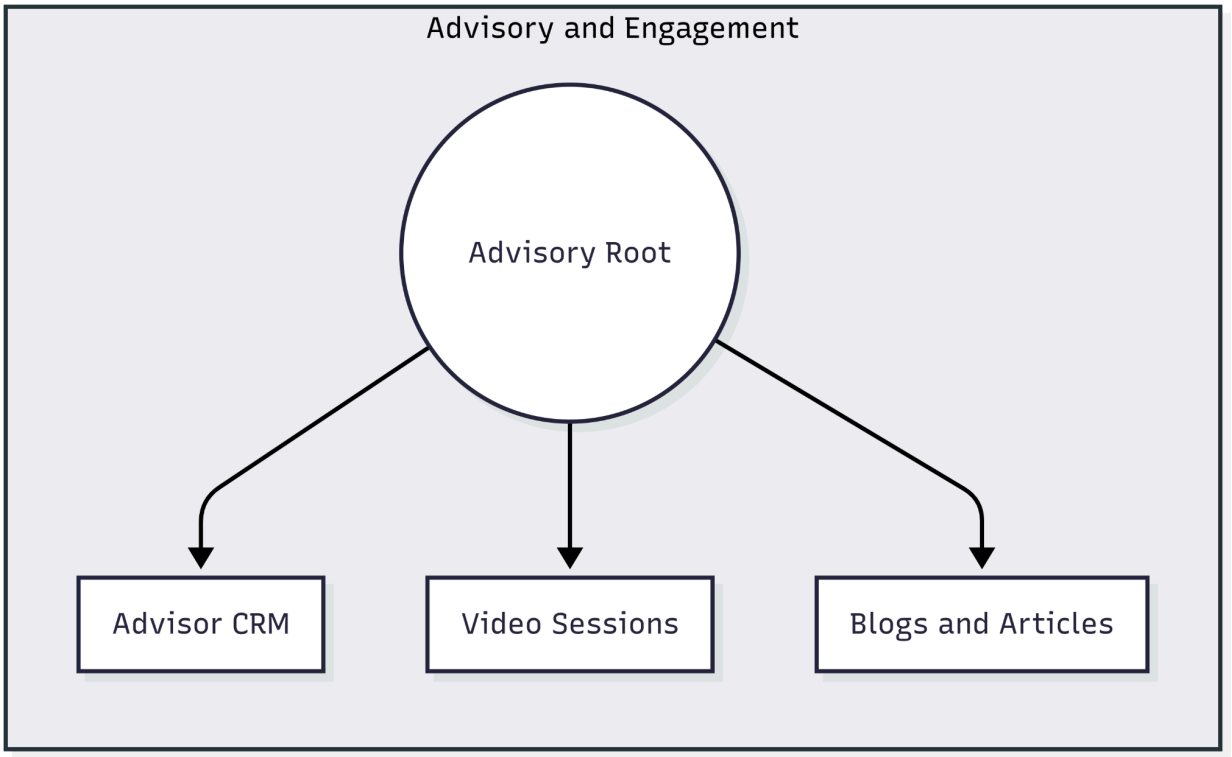


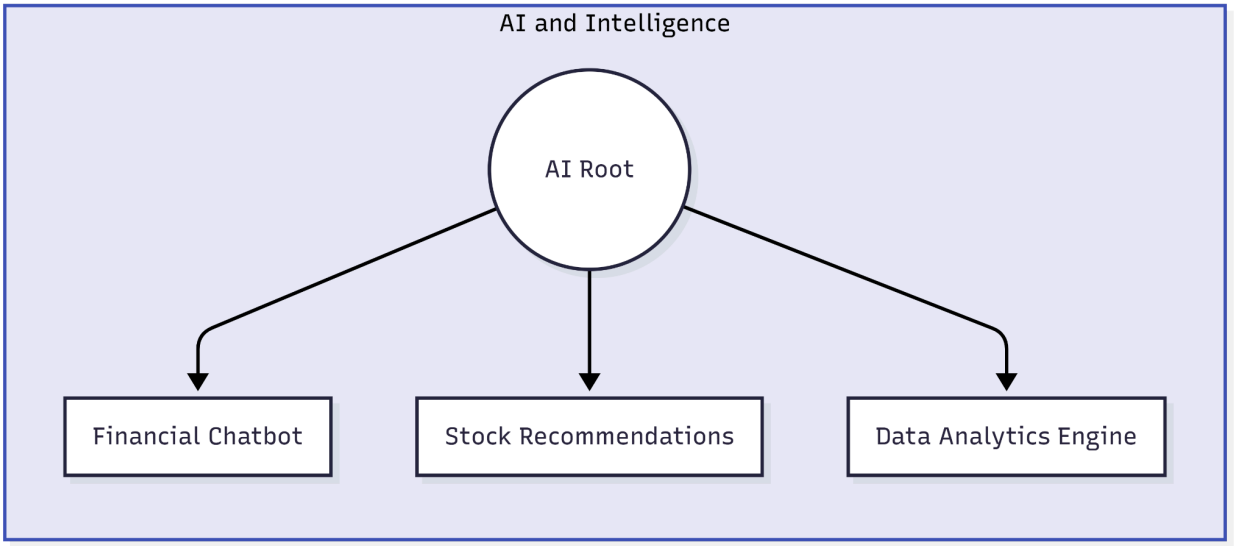

Identity and Access
IAM Root
Authentication
User Profiles


Financial Core
Fin Root
Bank Accounts
Cards
Expenses and Transactions


Asset and Wealth
Asset Root
Properties and Insurances
Fixed Deposits and Precious Holdings
Loans and EMIs


Investment
Invest Root
Stocks
Mutual Funds
Financial Goals

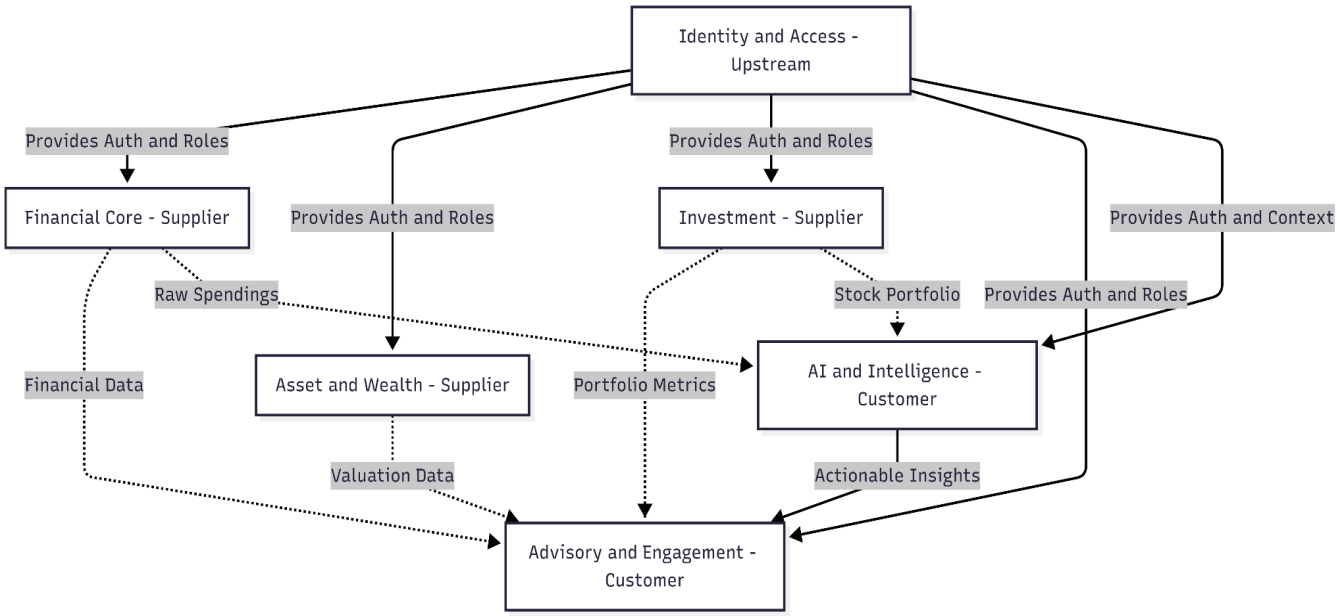

Advisory and Engagement
Advisory Root
Advisor CRM
Video Sessions
Blogs and Articles


AI and Intelligence
AI Root
Financial Chatbot
Stock Recommendations
Data Analytics Engine

B. Context Mappings

Integration between contexts relies primarily on a Customer-Supplier pattern. The system is designed defensively where core infrastructure supplies data to engagement and intelligence layers without bleeding domain logic.

- Upstream Identity: The Identity Context serves as the ultimate upstream dependency. Downstream contexts rely on the Shared Kernel of a JSON Web Token (JWT) payload to identify the user and authorize domain actions.
- Supplier Contexts: The Financial, Asset, and Investment contexts act as strictly isolated suppliers. They do not communicate with each other directly but store specialized data independently tied back to user identities.
- Customer Context: The Advisory and AI Contexts act as customers to the supplier contexts. When authorized via mapping collections, these contexts aggregate data from Financial, Asset, and Investment domains strictly for execution of Read operations to populate the Advisor Dashboard, generate PDF reports, and formulate AI context windows.



C. Domain Architecture: Entities, Value Objects, & Services

Each Bounded Context manages its distinct components. Entities possess strict identities, indicating they undergo structural state changes over time. Value Objects are immutable, defined purely by their attributes. Domain Services encapsulate

Bounded Context	Root & Child Entities	Value Objects	Domain Services
Identity & Access	Person (Root) User, Advisor, Admin	Credentials (Cryptographic Hash) ContactInfo (Email, Phone) RoleType (Enum)	AuthenticationService ProfileManagementService
Financial Core	Transaction (Root) BankAccount, Card, Expense EMI	Money (Amount, Currency Code) DateRange CategoryType	CategorizationService FinancialAnalyticsService
Asset & Wealth	Asset (Root) Property, Insurance, Loan FixedDeposit, PreciousHolding	ValuationAmount InterestRate PolicyDetails	AssetValuationService AmortizationCalculator
Investment	Portfolio (Root) Stock, MutualFund FinancialGoal	TickerSymbol RiskProfile NetAssetValue	MarketDataService GoalTrackingService
Advisory Model	AdvisorySession (Root) UserAdvisorLink Blog	VideoRoomURL ConsultationSchedule	MatchmakingService ReportGenerationService

Bounded Context	Root & Child Entities	Value Objects	Domain Services
AI Intelligence	AI_Interaction (Root) Recommendation	PromptPayload SentimentScore ConfidenceInterval	LLM_OrchestrationService PredictiveAnalysisService

D. Cardinality Ratios

The relationships defining the underlying schema dictate how context mappings are persisted physically within the document database.

Constraint Type	Relationship Map	Description
Inheritance / 1:1 Constraints	Person 1 : 1 (User | Advisor | Admin)	Implements strict sub-typing extension. A generalized persona is extended exclusively into one operational role per account.
Composition / 1:N Constraints	User 1 : N Expenses & Transactions	A user can have multiple expenses spanning their lifecycle.
	User 1 : N BankAccounts, Cards, EMIs	A user manages multiple instances of internal tracking instruments.
	User 1 : N Investments	A user owns multiple volatile assets spanning stocks and mutual funds tracking toward FinancialGoals.

Constraint Type	Relationship Map	Description
	User 1 : N Assets	A user owns multiple static assets encompassing property, insurances, fixed deposits, and precious holdings.
	Advisor 1 : N Blogs	An advisor maintains an exclusive authorship over published articles.
Aggregation / M:N Constraints	User M : N Advisor	Facilitated through intermediate mapping collections (managed by the Entity UserAdvisorLink), allowing a single user to engage multiple specialist advisors simultaneously while an advisor concurrently manages a broader client pool.

E. Domain Aggregates

Aggregates define a strong boundary around one or more entities to ensure transactional consistency and enforce strict business invariants (rules). Dependent entity data mutations must occur explicitly through the Aggregate Root.

1. The User Fin-Core Aggregate

- Aggregate Root: User
- Boundaries: BankAccounts, Cards, Expenses, Transactions, EMIs.
- Invariants: An Expense entity must rigidly link to an existing BankAccount or Card entity within the same user boundary. If the User root is deleted from the Identity Context, all boundaries within the Fin-Core aggregate must instantly undergo cascade-deletion to avoid orphaned transactional data.

2. The Wealth Portfolio Aggregate

- Aggregate Root: Portfolio (Bound strictly to the User profile)
- Boundaries: Stocks, MutualFunds, Properties, Insurances, FixedDeposits, PreciousHoldings.
- Invariants: The total net worth calculation forms an invariant that must consistently sum current valuations across all asset boundaries. Modifying an underlying asset (e.g., updating a property's market value) triggers a forced recalculation of the encompassing Portfolio aggregate state.

3. The Advisory Relationship Aggregate

- Aggregate Root: Advisor
- Boundaries: UserAdvisorLink (Relationship mappings), Assigned Clients.
- Invariants: Core access control parameters dictate that User data accessible to the Advisor root must be treated as structurally immutable (strictly Read-Only) proxy data. Furthermore, establishing this aggregate boundary requires bi-directional consent. A user cannot be injected into the advisor's client boundary without recorded explicit approval.

4. The Intelligence Aggregate

- Aggregate Root: AI_Interaction
- Boundaries: PromptPayload, Recommendation
- Invariants: An AI_Interaction must be permanently bound to a single User identity. Context injected into the prompts by the DRAKZ-1 middleware must be strictly read-only and cannot trigger mutations in the Fin-Core or Investment databases directly.